Convolution using NN's (excludes the bias term).

| | | |
|---|---|---|
| $x_{11}$ | $x_{12}$ | $x_{13}$ |
| $x_{21}$ | $x_{22}$ | $x_{23}$ |
| $x_{31}$ | $x_{32}$ | $x_{33}$ |

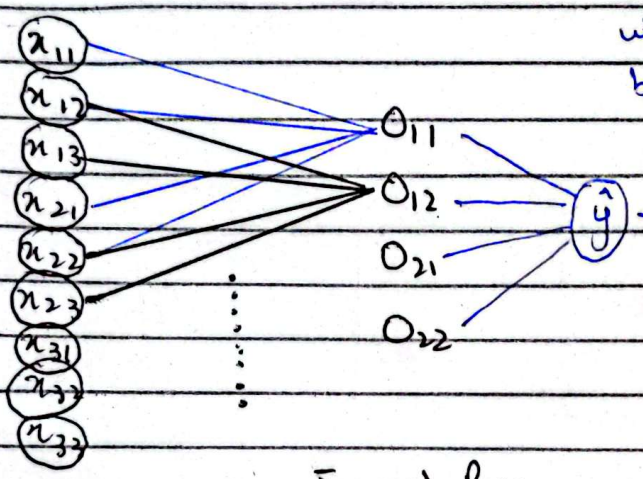
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| | |
|---|---|
| $f_{11}$ | $f_{12}$ |
| $f_{21}$ | $f_{22}$ |

$=$

| | |
|---|---|
| $O_{11}$ | $O_{12}$ |
| $O_{21}$ | $O_{22}$ |

↳ Note: The convolution kernel will provide the weights to be learned.




Forward Pass

$$O_{11} = f_{11}x_{11} + f_{12}x_{12} + f_{21}x_{21} + f_{22}x_{22}$$
$$O_{12} = f_{11}x_{12} + f_{12}x_{13} + f_{21}x_{22} + f_{22}x_{23}$$
$$O_{21} = f_{11}x_{21} + f_{12}x_{22} + f_{21}x_{31} + f_{22}x_{32}$$
$$O_{22} = f_{11}x_{22} + f_{12}x_{23} + f_{21}x_{32} + f_{22}x_{33}$$

Back Prop

$$w_i' = w_i - \alpha \boxed{\frac{\partial L}{\partial w_i}}$$

↑ TWS is unknown

→ weight updation expression

$f_{ij}$ → from kernel / filter

(please note in example above → $f_{ij}$ are weight params)

since # of weights = # of partial derivatives we can represent $\frac{\partial L}{\partial w_i}$ using a kernel

↳

| | |
|---|---|
| $\frac{\partial L}{\partial f_{11}}$ | $\frac{\partial L}{\partial f_{12}}$ |
| $\frac{\partial L}{\partial f_{21}}$ | $\frac{\partial L}{\partial f_{22}}$ |

**Next** → calculate the partial derv of Loss w.r.t weights.

Intuition: notice from the diagram how a change in $f_{11}$ for example, will end up changing all of the $O_{ij}$ values.
→ This happens because $f_{11}$ appears in all of the eqns for the O's → this would cause $\hat{y}$ and Loss to change as well.

↓ now lets try visualizing this via backprop.

$$\frac{\partial L}{\partial f_{11}} = \left(\left(\frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial O_{11}}\right) \cdot \frac{\partial O_{11}}{\partial f_{11}}\right) + \left(\frac{\partial L}{\partial O_{12}} \cdot \frac{\partial O_{12}}{\partial f_{11}}\right) + \frac{\partial L}{\partial O_{21}} \cdot \frac{\partial O_{21}}{\partial f_{11}} + \frac{\partial L}{\partial O_{22}} \cdot \frac{\partial O_{22}}{\partial f_{11}}$$

($\frac{\partial L}{\partial O_{11}}$ to cater for multiple pathways)

(now use the equations from Forward Pass to simplify terms in Back Prop).

$\frac{\partial O_{11}}{\partial f_{11}} = x_{11}$ since $O_{11} = x_{11} f_{11} + x_{12} f_{12} + x_{21} f_{21} + x_{22} f_{22}$.

$\frac{\partial L}{\partial f_{11}}$ becomes → $\frac{\partial L}{\partial O_{11}} \cdot \frac{\partial O_{11}}{\partial f_{11}} + \cdots$

∴ ↳ (simplifying)

$$\frac{\partial L}{\partial O_{11}} \cdot x_{11} + \frac{\partial L}{\partial O_{12}} \cdot x_{12} + \frac{\partial L}{\partial O_{21}} \cdot x_{21} + \frac{\partial L}{\partial O_{22}} \cdot x_{22}$$

now repeat for all other partial derivatives. (simplifying).

$$\frac{\partial L}{\partial f_{11}} = \frac{\partial L}{\partial O_{11}} \cdot x_{11} + \frac{\partial L}{\partial O_{12}} \cdot x_{12} + \frac{\partial L}{\partial O_{21}} \cdot x_{21} + \frac{\partial L}{\partial O_{22}} \cdot x_{22}$$

$$\frac{\partial L}{\partial f_{12}} = \frac{\partial L}{\partial O_{11}} x_{12} + \frac{\partial L}{\partial O_{12}} \cdot x_{13} + \frac{\partial L}{\partial O_{21}} \cdot x_{22} + \frac{\partial L}{\partial O_{22}} \cdot x_{23}$$

$$\frac{\partial L}{\partial f_{21}} = \frac{\partial L}{\partial O_{11}} x_{21} + \frac{\partial L}{\partial O_{12}} x_{22} + \frac{\partial L}{\partial O_{21}} x_{31} + \frac{\partial L}{\partial O_{22}} x_{32}$$

$$\frac{\partial L}{\partial f_{22}} = \frac{\partial L}{\partial O_{11}} x_{22} + \frac{\partial L}{\partial O_{12}} x_{23} + \frac{\partial L}{\partial O_{21}} x_{32} + \frac{\partial L}{\partial O_{22}} x_{33}$$

(now rewrite the equations on previous page in matrix form)

notice how across columns in eqns on prev page, the $\frac{\partial L}{\partial o_{ij}}$ term remains constant and only the weight values ($x_{ij}$) change.

$$\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \cdot \frac{\partial L}{\partial o_{11}} + \begin{bmatrix} x_{12} & x_{13} \\ x_{22} & x_{23} \end{bmatrix} \cdot \frac{\partial L}{\partial o_{12}} + \begin{bmatrix} x_{21} & x_{22} \\ x_{31} & x_{32} \end{bmatrix} \frac{\partial L}{\partial o_{21}} + \begin{bmatrix} x_{22} & x_{23} \\ x_{32} & x_{33} \end{bmatrix} \frac{\partial L}{\partial o_{22}}$$

notice how solving the above expr would result in $\rightarrow$ $\begin{bmatrix} \frac{\partial L}{\partial f_{11}} & \frac{\partial L}{\partial f_{12}} \\ \frac{\partial L}{\partial f_{21}} & \frac{\partial L}{\partial f_{22}} \end{bmatrix}$ $\rightarrow$ notice how this is equ to

$$\text{conv}\left(\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}, \begin{bmatrix} \frac{\partial L}{\partial o_{11}} & \frac{\partial L}{\partial o_{12}} \\ \frac{\partial L}{\partial o_{21}} & \frac{\partial L}{\partial o_{22}} \end{bmatrix}\right)$$

now we can use gradient descent to update the weights.

$$\begin{bmatrix} f_{11}' & f_{12}' \\ f_{21}' & f_{22}' \end{bmatrix} = \begin{bmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{bmatrix} - \alpha \cdot \begin{bmatrix} \frac{\partial L}{\partial f_{11}} & \frac{\partial L}{\partial f_{12}} \\ \frac{\partial L}{\partial f_{21}} & \frac{\partial L}{\partial f_{22}} \end{bmatrix}$$

($\alpha \rightarrow$ L.R)

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Important Point: in a multi layer CNN, input $X$ is usually the output of a previous layer, so in order to keep propogating the loss backward, we need gradient of $L$ w.r.t $X$ as well $\rightarrow$ i.e $\frac{\partial L}{\partial X}$ becomes the error signal for previous layer.

Lets compute this as well. $\rightarrow$

$$\frac{\partial L}{\partial x_{11}} = \frac{\partial L}{\partial o_{11}} \cdot f_{11} + \frac{\partial L}{\partial o_{12}} \cdot 0 + \frac{\partial L}{\partial o_{21}} \cdot 0 + \frac{\partial L}{\partial o_{22}} \cdot 0$$

$$\frac{\partial L}{\partial x_{12}} = \frac{\partial L}{\partial o_{11}} \cdot f_{12} + \frac{\partial L}{\partial o_{12}} \cdot f_{11} + \frac{\partial L}{\partial o_{21}} \cdot 0 + \frac{\partial L}{\partial o_{22}} \cdot 0$$

$$\frac{\partial L}{\partial x_{13}} = \frac{\partial L}{\partial o_{11}} \cdot 0 + \frac{\partial L}{\partial o_{12}} \cdot f_{12} + \frac{\partial L}{\partial o_{21}} \cdot 0 + \frac{\partial L}{\partial o_{22}} \cdot 0$$

$$\frac{\partial L}{\partial x_{21}} = \frac{\partial L}{\partial o_{11}} \cdot f_{21} + \frac{\partial L}{\partial o_{12}} \cdot 0 + \frac{\partial L}{\partial o_{21}} f_{11} + \frac{\partial L}{\partial o_{22}} \cdot 0$$

$$\frac{\partial L}{\partial x_{33}} = \frac{\partial L}{\partial o_{11}} \cdot 0 + \frac{\partial L}{\partial o_{12}} \cdot 0 + \frac{\partial L}{\partial o_{21}} \cdot 0 + \frac{\partial L}{\partial o_{22}} \cdot f_{22}$$

↓ this computation can also be obtained by full convolution.

| | | | |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | $\frac{\partial L}{\partial o_{11}}$ | $\frac{\partial L}{\partial o_{12}}$ | 0 |
| 0 | $\frac{\partial L}{\partial o_{21}}$ | $\frac{\partial L}{\partial o_{22}}$ | 0 |
| 0 | 0 | 0 | 0 |

$*$

| | |
|---|---|
| $f_{22}$ | $f_{21}$ |
| $f_{12}$ | $f_{11}$ |

↓ 180° rotation (hori + vert flip)

$\longrightarrow$

$$\begin{bmatrix} \frac{\partial L}{\partial x_{11}} & \frac{\partial L}{\partial x_{12}} & \frac{\partial L}{\partial x_{13}} \\ \frac{\partial L}{\partial x_{21}} & \frac{\partial L}{\partial x_{22}} & \frac{\partial L}{\partial x_{23}} \\ \frac{\partial L}{\partial x_{31}} & \frac{\partial L}{\partial x_{32}} & \frac{\partial L}{\partial x_{33}} \end{bmatrix}$$

↓ now this will flow into previous layer as gradient of loss w.r.t output and play the same role that was playing in the current layer.

$$\leftarrow \begin{bmatrix} \frac{\partial L}{\partial o_{11}} & \frac{\partial L}{\partial o_{12}} \\ \frac{\partial L}{\partial o_{21}} & \frac{\partial L}{\partial o_{22}} \end{bmatrix}$$